

1. Input

Section, materials and resistance basis exactly as analysed.

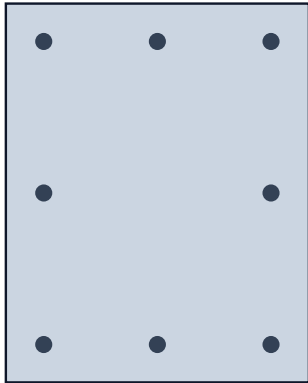
Identity

Project	ACI-EB 01 - Rectangle 400 x 500 m.	Section	Benchmark 01 - Rectangle 400 x 50.
Calculation profile	ACI 318-19(22) — Whitney equivalence.	Mechanics	Equivalent rectangular stress block
Resistance profile	ACI CODE-318-19 (Reapproved 202.	Profile edition	2019 (Reapproved 2022)
Verification status	draft	Method id	aci-318-19-22-global-strength-reduct.
Units	mm · N · MPa; forces reported in kN.	Sign convention	Compression positive; $M_x = \Sigma F \cdot (y - y_c)$.

Section

Cross-section (mm, about the net concrete centroid)

8 bars



Section properties

Gross concrete area A_g	200,000 mm ²
Reinforcement area A_s	3,200 mm ²
Reinforcement ratio ρ	1.600 %
Bar count	8
Net centroid	(0.0, 0.0) mm
Bounding box	400.0 × 500.0 mm
Solids / holes	1 / 0
Perimeter	1,800.0 mm

Materials

Name	Concrete C40	Standard	ACI318
f_{ck}	40.0 MPa	Model	aci-whitney-block
ϵ_{cu}	0.00300	α	0.8500
E_c	29,725 MPa	Tension	ignored
Name	Rebar fy 420	Standard	ACI318
f_y	420.0 MPa	E_s	200,000 MPa
Model	elastic-perfectly-plastic	ϵ_y	0.002100
ϵ_u	not declared	Grades used	1

Design resistance basis

Format	Global resultant strength-reduction f.	Transverse reinforcement	other
ϕ compression (ties)	0.650	ϕ compression (spiral)	0.750
ϕ tension	0.900	$\epsilon_{t,limit}$	0.00510
Transition rule	yield-plus-strain	Axial cap	0.800 × factored compression pole

Sampling and numerical evidence

Surface points	4905	Directions	120
Stations	52	Direction tolerance	0.750 %
Worst interpolation error	0.7347 %	Tolerance reached	yes
Concrete discretization	exact polygon clipping	Strain domain	concrete-pivot-ultimate

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is the code-equivalent rectangular block over $a = \beta_1 \cdot c$, obtained by exact polygon clipping. There is no concrete integration mesh, and the stress between a and c is zero.

Reinforcement schedule

#	Bar id	(mm)	X (mm)	Y (mm)	A_s (mm ²)	Steel
1	1	22.6	-150.0	-200.0	400.0	Rebar fy 420

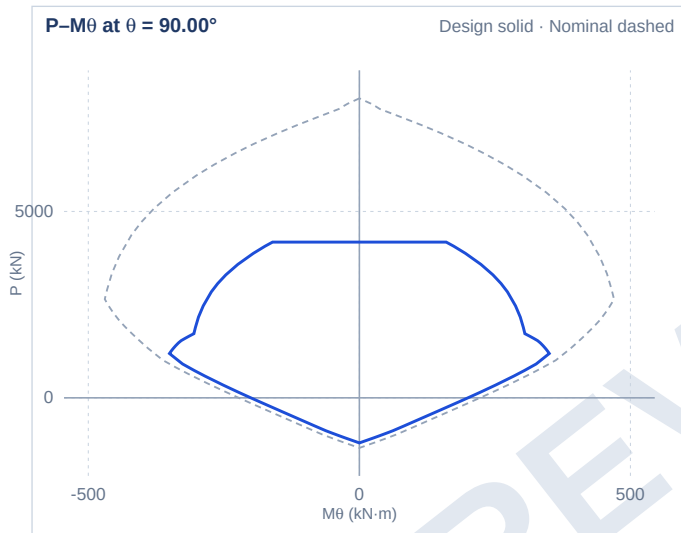
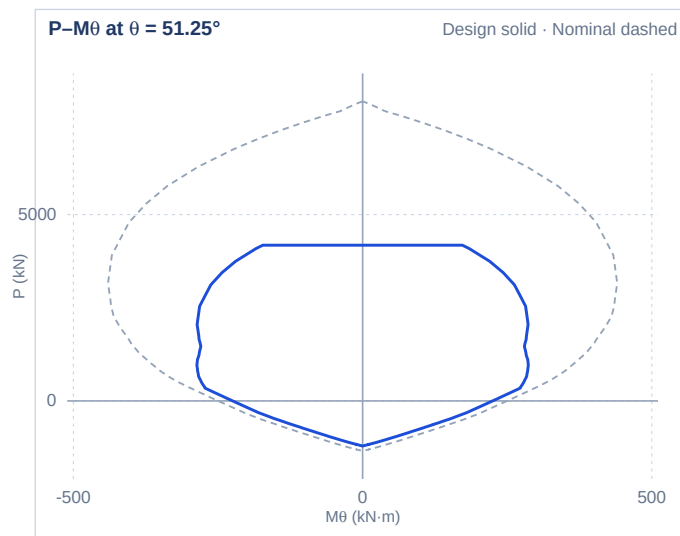
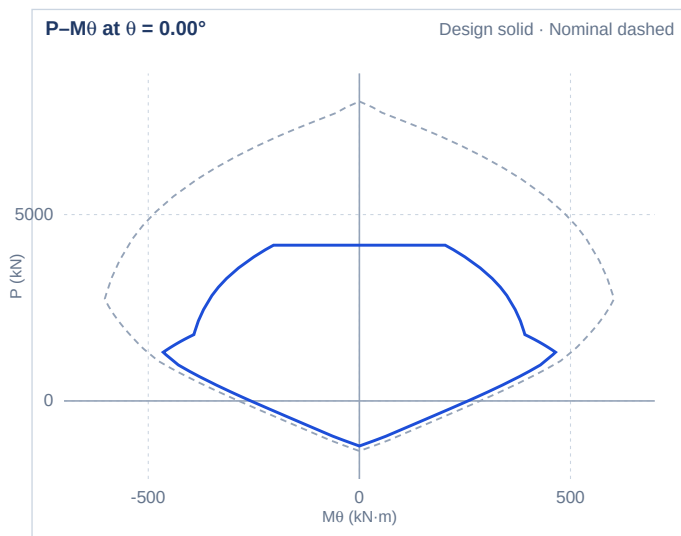
#	Bar id	(mm)	X (mm)	Y (mm)	As (mm ²)	Steel
2	2	22.6	0.0	-200.0	400.0	Rebar fy 420
3	3	22.6	150.0	-200.0	400.0	Rebar fy 420
4	4	22.6	150.0	0.0	400.0	Rebar fy 420
5	5	22.6	150.0	200.0	400.0	Rebar fy 420
6	6	22.6	0.0	200.0	400.0	Rebar fy 420
7	7	22.6	-150.0	200.0	400.0	Rebar fy 420
8	8	22.6	-150.0	0.0	400.0	Rebar fy 420

PREVIEW

2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.

Maximum design axial resistance applied at 4178.3 kN; it appears as the flat top of each diagram.



3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	P _u (kN)	M _{ux} (kN·m)	M _{uy} (kN·m)	θ _L (°)	φ	Class	UR	Verdict
1	Audit ray 55% - NA 26.93 d.	661.2	97.5	121.5	51.3	0.694	transition	0.550	ADEQUATE
2	Cardinal audit 85% - NA 0 d.	2178.8	0.0	242.0	90.0	0.650	compression-co.	0.850	ADEQUATE
3	Cardinal audit 85% - NA 90 .	1606.2	330.7	0.0	0.0	0.650	compression-co.	0.850	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

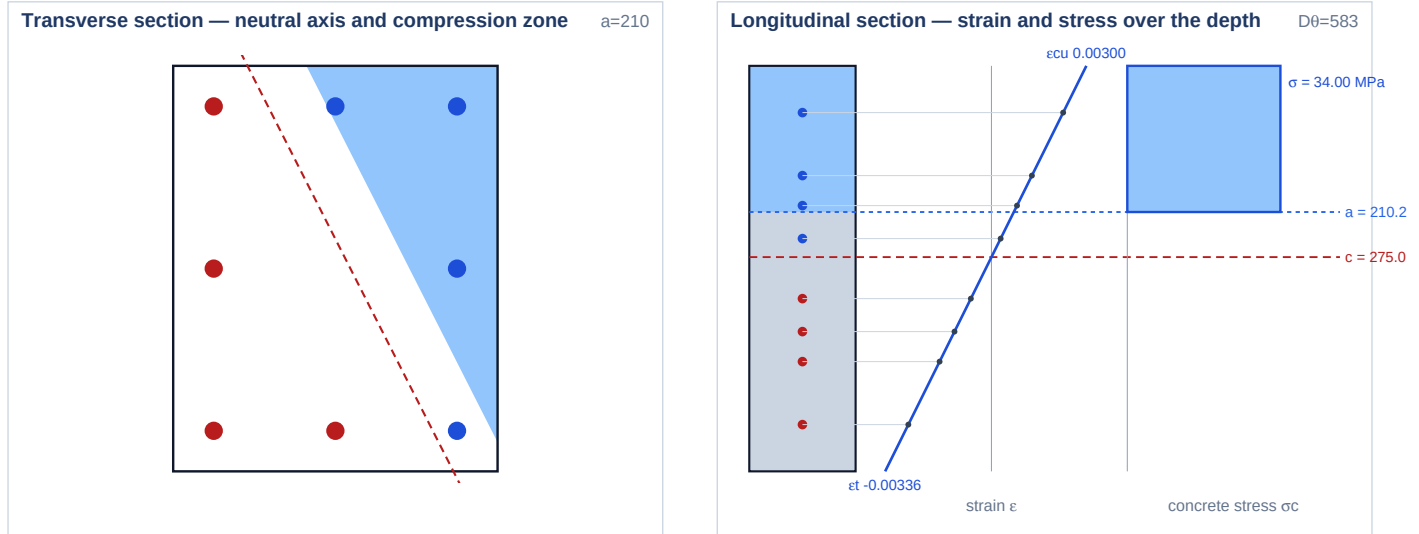
4.1 Audit ray 55% - NA 26.93 deg, c/D 0.55

$P_u = 661.2 \text{ kN}$ · $M_{ux} = 97.5 \text{ kN}\cdot\text{m}$ · $M_{uy} = 121.5 \text{ kN}\cdot\text{m}$ · $\theta_L = 51.25^\circ$

ADEQUATE · utilization 0.5500 · fixed-P diagnostic 0.5491

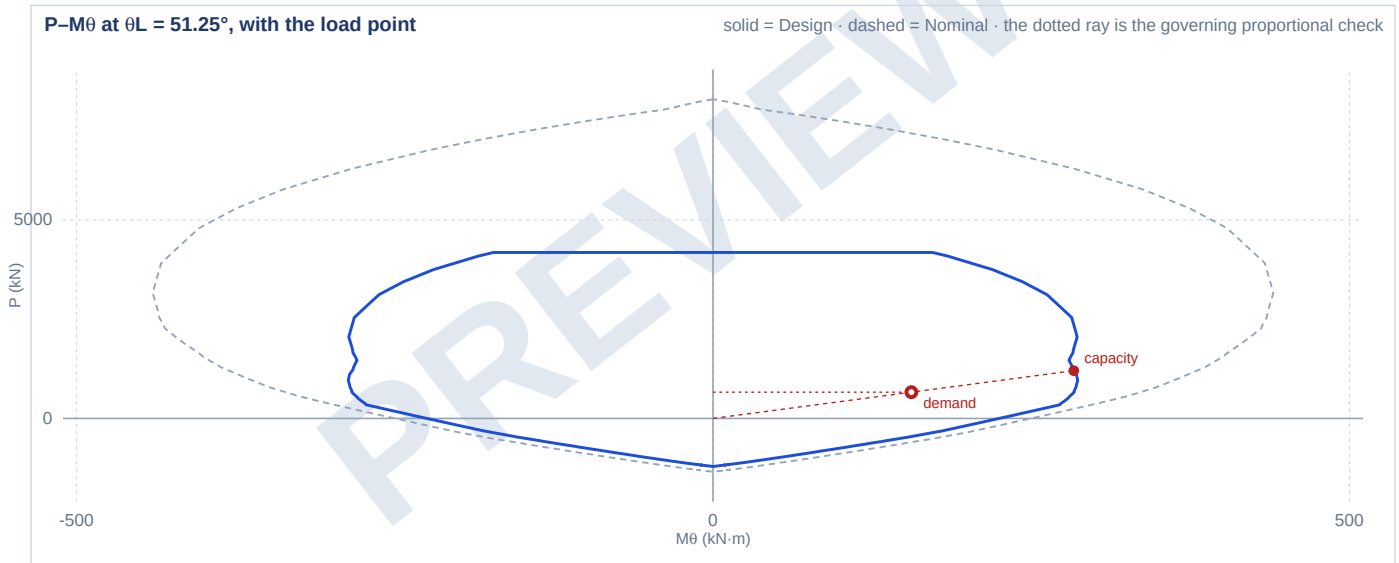
Factored ULS demand checked by the equivalent-block proportional-ray solver.

Section state



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Interaction diagram through the demand direction



Concrete and resultants

Block area $A_{c,blk}$	54,702.0 mm ²	Block centroid	(121.4, 95.3) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	1,859.9 kN
Concrete M_{cx}	177.3 kN·m	Concrete M_{cy}	225.8 kN·m
Nominal P_n	1,732.5 kN	Nominal M_{nx}	255.6 kN·m
Nominal M_{ny}	318.5 kN·m	ϕ	0.6940
Design ϕP_n	1,202.3 kN	Design ϕM_{nx}	177.4 kN·m
Design ϕM_{ny}	221.0 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	A_s (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	515.8	-2.627e-3	-420.00	-168.00	no
2	0.0	-200.0	400.0	382.1	-1.168e-3	-233.70	-93.48	no
3	150.0	-200.0	400.0	248.4	2.904e-4	58.09	23.23	no
4	150.0	0.0	400.0	157.8	1.279e-3	255.71	88.68	yes

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	150.0	200.0	400.0	67.2	2.267e-3	420.00	154.40	yes
6	0.0	200.0	400.0	201.0	8.077e-4	161.55	51.02	yes
7	-150.0	200.0	400.0	334.7	-6.512e-4	-130.24	-52.10	no
8	-150.0	0.0	400.0	425.3	-1.639e-3	-327.86	-131.15	no

Solver and admissibility evidence

Neutral-axis angle θ	26.9290°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β1	0.7643
Projected depth Dθ	583.07 mm	Controlling εt	0.002627
Controlling bar	1	Classification	transition
Axial cap governs	no	Component residual	1.02e-10
Strain admissibility	admissible	Governing utilization	0.5500

4.2 Cardinal audit 85% - NA 0 deg, c/D 0.55

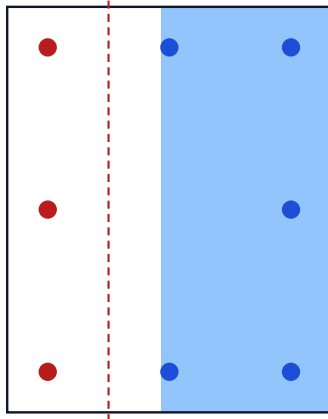
$P_u = 2178.8 \text{ kN}$ · $M_{ux} = 0.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 242.0 \text{ kN}\cdot\text{m}$ · $\theta L = 90.00^\circ$

ADEQUATE · utilization **0.8500** · fixed-P diagnostic **0.8163**

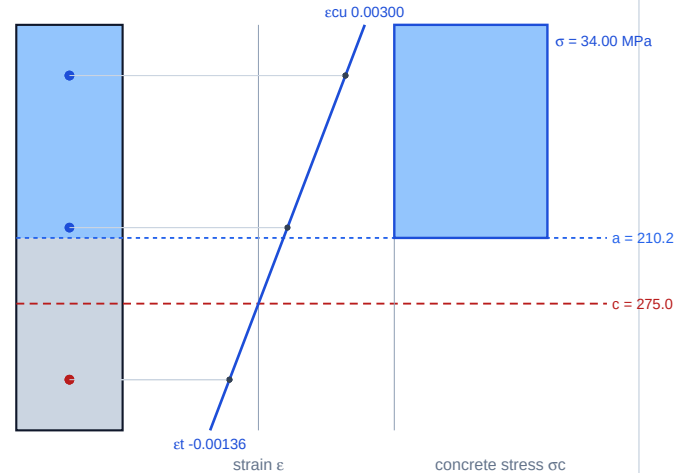
Factored ULS demand checked by the equivalent-block proportional-ray solver.

Section state

Transverse section — neutral axis and compression zone $a=210$



Longitudinal section — strain and stress over the depth $D\theta=400$

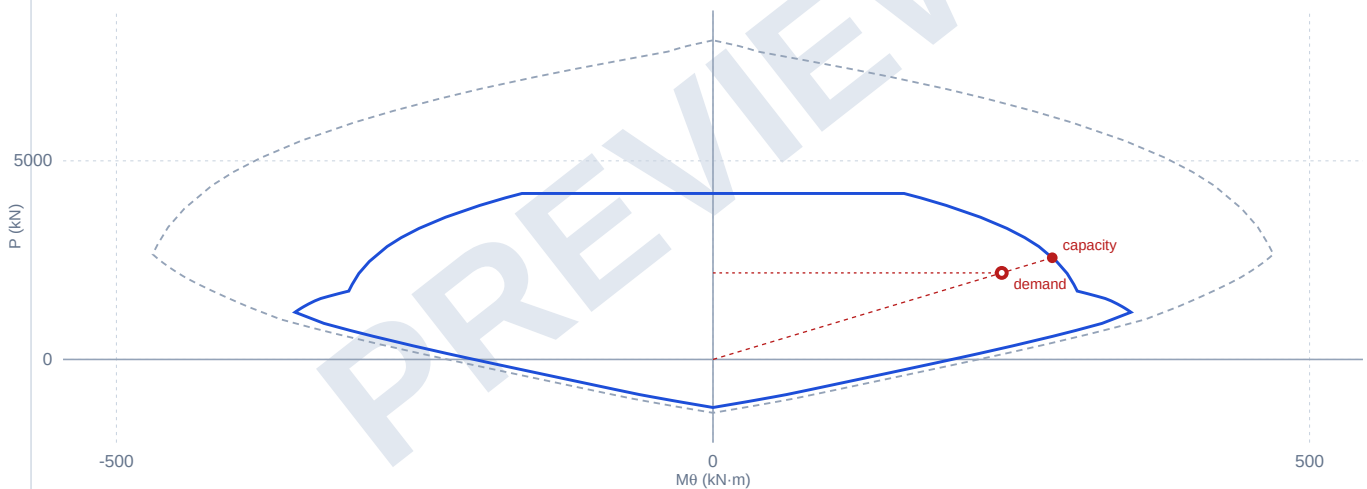


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Interaction diagram through the demand direction

P-M θ at $\theta L = 90.00^\circ$, with the load point

solid = Design · dashed = Nominal · the dotted ray is the governing proportional check



Concrete and resultants

Block area $A_{c,blk}$	105,089.3 mm ²	Block centroid	(94.9, -0.0) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	3,573.0 kN
Concrete M_{cx}	-0.0 kN·m	Concrete M_{cy}	339.1 kN·m
Nominal P_n	3,943.6 kN	Nominal M_{nx}	-0.0 kN·m
Nominal M_{ny}	438.1 kN·m	ϕ	0.6500
Design ϕP_n	2,563.3 kN	Design ϕM_{nx}	-0.0 kN·m
Design ϕM_{ny}	284.7 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
2	0.0	-200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
3	150.0	-200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
4	150.0	0.0	400.0	50.0	2.455e-3	420.00	154.40	yes

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
7	-150.0	200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
8	-150.0	0.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no

Solver and admissibility evidence

Neutral-axis angle θ	0.0000°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β1	0.7643
Projected depth Dθ	400.00 mm	Controlling εt	0.000818
Controlling bar	1	Classification	compression-controlled
Axial cap governs	no	Component residual	-1.75e-10
Strain admissibility	admissible	Governing utilization	0.8500

4.3 Cardinal audit 85% - NA 90 deg, c/D 0.55

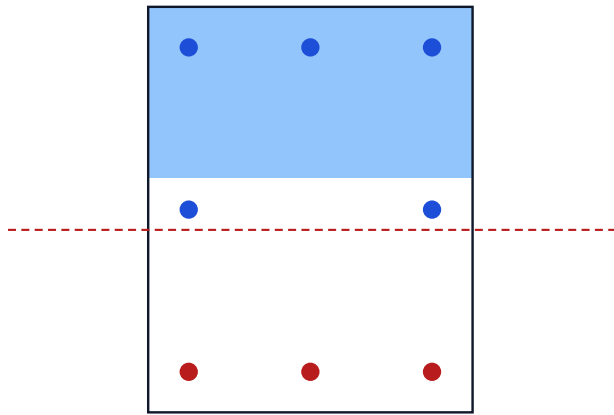
$P_u = 1606.2 \text{ kN}$ · $M_{ux} = 330.7 \text{ kN}\cdot\text{m}$ · $M_{uy} = 0.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 0.00^\circ$

ADEQUATE · utilization **0.8500** · fixed-P diagnostic **0.7840**

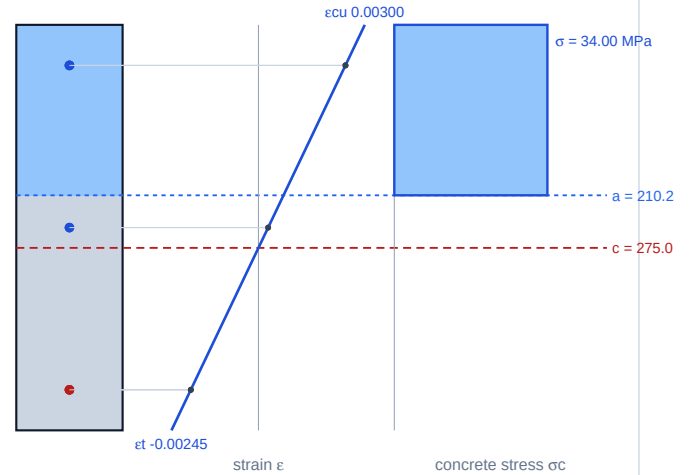
Factored ULS demand checked by the equivalent-block proportional-ray solver.

Section state

Transverse section — neutral axis and compression zone $a=210$



Longitudinal section — strain and stress over the depth $D\theta=500$

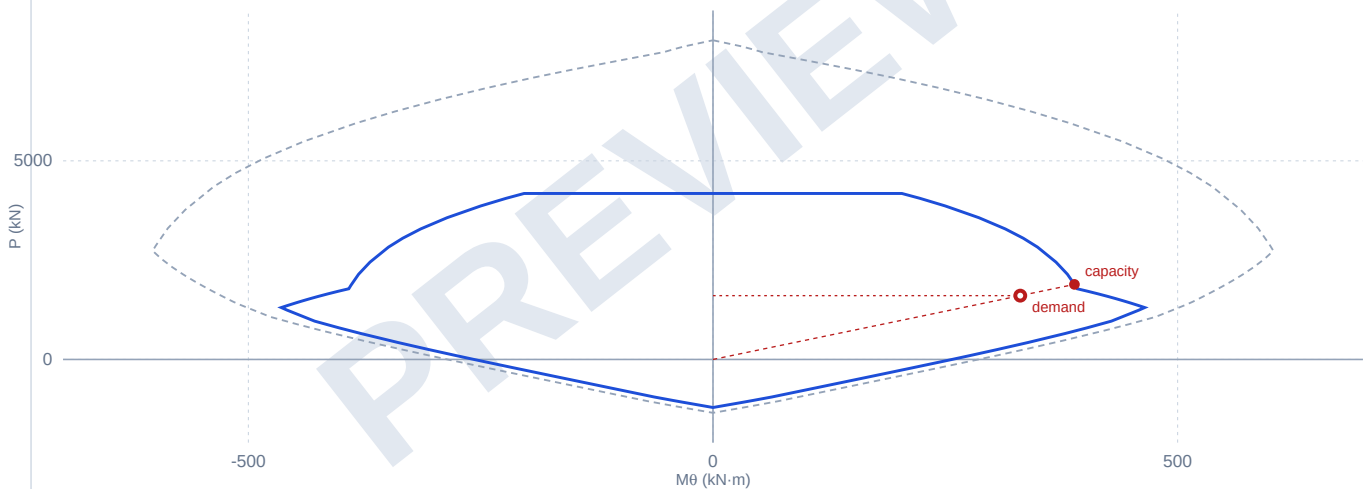


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Interaction diagram through the demand direction

P-M θ at $\theta_L = 0.00^\circ$, with the load point

solid = Design · dashed = Nominal · the dotted ray is the governing proportional check



Concrete and resultants

Block area $A_{c,blk}$	84,071.4 mm ²	Block centroid	(0.0, 144.9) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	2,858.4 kN
Concrete M_{cx}	414.2 kN·m	Concrete M_{cy}	0.0 kN·m
Nominal P_n	2,907.1 kN	Nominal M_{nx}	598.5 kN·m
Nominal M_{ny}	0.0 kN·m	ϕ	0.6500
Design ϕP_n	1,889.6 kN	Design ϕM_{nx}	389.0 kN·m
Design ϕM_{ny}	0.0 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
2	0.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
3	150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
4	150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
7	-150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
8	-150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no

Solver and admissibility evidence

Neutral-axis angle θ	90.0000°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β1	0.7643
Projected depth Dθ	500.00 mm	Controlling εt	0.001909
Controlling bar	1	Classification	compression-controlled
Axial cap governs	no	Component residual	-6.55e-11
Strain admissibility	admissible	Governing utilization	0.8500